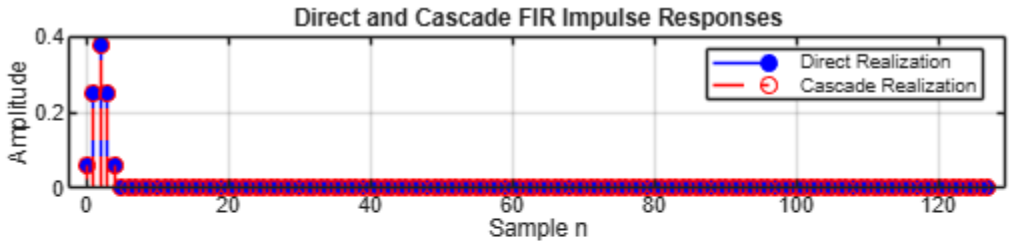
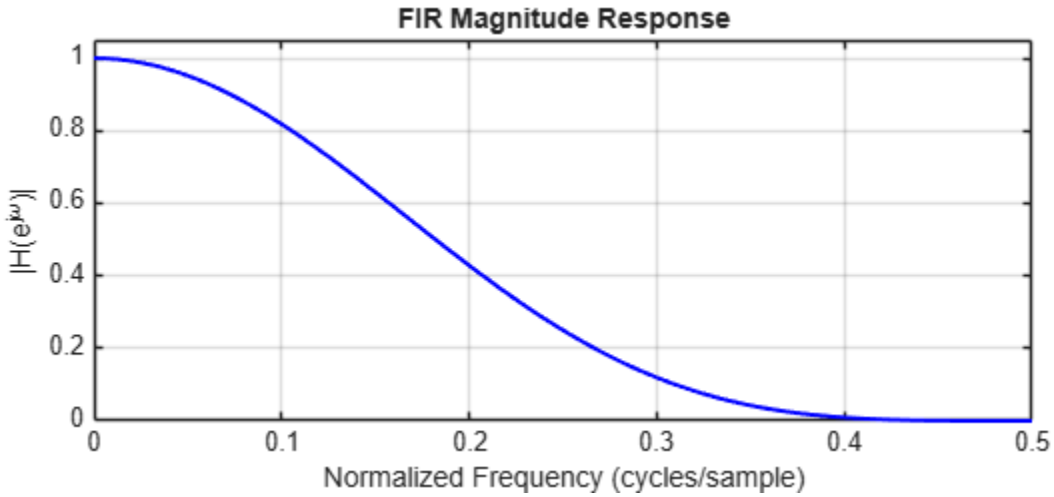
```
clc;
clear;
close all;

% FIR Filter: Direct and Cascade Realizations

% 1. FIR Filter Coefficients
b1 = [0.25 0.5 0.25];
b2 = [0.25 0.5 0.25];

% Direct realization coefficients
b_direct = conv(b1, b2);

% 2. Impulse Signal
N = 128;
x = [1 zeros(1,N-1)];

% 3. Direct Realization
y_direct = filter(b_direct, 1, x);

% 4. Cascade Realization
y_stage1 = filter(b1, 1, x);
y_cascade = filter(b2, 1, y_stage1);

% 5. GRAPH 1: Impulse Responses
figure(1);

subplot(2,1,1);
stem(0:N-1, y_direct, 'b', 'filled');
hold on;
stem(0:N-1, y_cascade, 'r--');
grid on;

title('Direct and Cascade FIR Impulse Responses');
xlabel('Sample n');
ylabel('Amplitude');

legend('Direct Realization','Cascade Realization');

% 6. GRAPH 2: Magnitude Response
nfft = 1024;

H = fft(b_direct, nfft);
H = H(1:nfft/2+1);

f = linspace(0, 0.5, nfft/2+1);
magnitude = abs(H);

figure(2);

plot(f, magnitude, 'b', 'LineWidth', 1.5);
grid on;
```

```
title('FIR Magnitude Response');
xlabel('Normalized Frequency (cycles/sample)');
ylabel('|H(e^{j\omega})|');

xlim([0 0.5]);
ylim([0 1.05]);
```





Published with MATLAB® R2026b